ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-2 **SUMMER SEMESTER, 2022-2023**

DURATION: 40 MINUTES **FULL MARKS: 15**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. The Taylor series for a function $f(x)$ is — 6

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \dots$$ (CO1) (PO2)

Using the Taylor series (with at least the first 4 terms), derive the Maclaurin series of —

(i) $\ln(1+x)$

(ii) $\ln(1-x)$

(iii) $\ln\left(\frac{1+x}{1-x}\right)$.

Use all three series derived above to approximate the value of $\ln(2)$, and comment on their accuracy.
[**Hint:** Feel free to reuse a previously derived answer in response to another question if applicable.]

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$, and $\frac{d}{dx}(x^n) = nx^{n-1}$.

(i) Now, for $f(x) = \ln(1+x)$, let's calculate the derivatives at the point $x = 0$.

$f(x) = \ln(1+x)$, $f(0) = \ln(1+0) = \ln(1) = 0$; $f'(x) = \frac{1}{1+x}$, $f'(0) = \frac{1}{1+0} = 1$;

$f''(x) = -\frac{1}{(1+x)^2}$, $f''(0) = -\frac{1}{(1+0)^2} = -1$; $f'''(x) = \frac{2}{(1+x)^3}$, $f'''(0) = \frac{2}{(1+0)^3} = 2$;

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots$$
$$\Rightarrow f(h) = 0 + h - \frac{h^2}{2!} + \frac{2h^3}{3!} + \dots$$
$$\Rightarrow f(x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$
$$\Rightarrow \ln(1+x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$
$$\Rightarrow \ln(1+x) = x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

(ii) **Short version:** Here,

$$\begin{aligned} f(x) &= \ln(1-x) \\ &= \ln(1+(-x)) \\ &= (-x) - \frac{(-x)^2}{2!} + \frac{2(-x)^3}{3!} + \dots \qquad \text{[using the Maclaurin series of } \ln(1+x) \text{ from (i)]} \\ &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots \end{aligned}$$

Long version: Now, for $f(x) = \ln(1-x)$, let's calculate the derivatives at the point $x = 0$.

$f(x) = \ln(1-x)$, $f(0) = \ln(1-0) = \ln(1) = 0$; $f'(x) = -\frac{1}{1-x}$, $f'(0) = -\frac{1}{1-0} = -1$;
$f''(x) = -\frac{1}{(1-x)^2}$, $f''(0) = -\frac{1}{(1-0)^2} = -1$;
$f'''(x) = -\frac{2}{(1-x)^3}$, $f'''(0) = -\frac{2}{(1-0)^3} = -2$;

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots$$
$$\Rightarrow f(h) = 0 - h - \frac{h^2}{2!} - \frac{2h^3}{3!} + \dots$$
$$\Rightarrow f(x) = 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$
$$\Rightarrow \ln(1-x) = 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$
$$\Rightarrow \ln(1-x) = -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

We know, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$.

So,

$$\begin{aligned}\ln\left(\frac{1+x}{1-x}\right) &= \ln(1+x) - \ln(1-x) \\ &= \left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots\right) - \left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots\right) \\ &= 2x + \frac{4x^3}{3!} + \dots\end{aligned}$$

Approximating $\ln(2)$**:** The true value is $\ln(2) = 0.69315$ (to 5 decimal places). We now use each of the three series to estimate it.

(i) For $\ln(1+x)$, we set $1+x=2$, i.e. $x=1$:

$$\begin{aligned}\ln(2) &\approx 1 - \frac{1^2}{2!} + \frac{2\cdot 1^3}{3!} \\ &= 1 - 0.5 + 0.33333 \\ &= 0.83333\end{aligned}$$

This gives an absolute error of $|0.83333 - 0.69315| = 0.14019$.

(ii) For $\ln(1-x)$, we set $1-x=2$, i.e. $x=-1$:

$$\begin{aligned}\ln(2) &\approx -(-1) - \frac{(-1)^2}{2!} - \frac{2\cdot(-1)^3}{3!} \\ &= 1 - 0.5 + 0.33333 \\ &= 0.83333\end{aligned}$$

This gives the same absolute error of 0.14019, which is expected since (i) and (ii) are the same series evaluated at the same magnitude of x.

(iii) For $\ln\left(\frac{1+x}{1-x}\right)$, we set $\frac{1+x}{1-x} = 2$, which gives $1+x = 2(1-x) \Rightarrow 3x = 1 \Rightarrow x = \frac{1}{3}$:

$$\begin{aligned}\ln(2) &\approx 2\left(\frac{1}{3}\right) + \frac{4}{3!}\left(\frac{1}{3}\right)^3 \\ &= 0.66667 + 0.02469 \\ &= 0.69136\end{aligned}$$

This gives an absolute error of only $|0.69136 - 0.69315| = 0.00179$.

Although all three series use the same number of terms, the third series $\ln\left(\frac{1+x}{1-x}\right)$ is far more accurate (~ 0.002 error versus ~ 0.14 for the other two). This is because subtracting the two logarithm series cancels all the even-powered terms, leaving only odd powers, and we evaluate it at $x = \frac{1}{3}$, which is much smaller than the $|x| = 1$ used for the first two series. A smaller evaluation point makes the neglected higher-order terms shrink rapidly, so the truncated series converges much faster. **Rubric:**

- 2 points for the correct series of $\ln(1+x)$.
- 2 points for the correct series of $\ln(1-x)$.
- 1 point for the correct series of $\ln\left(\frac{1+x}{1-x}\right)$.
- 1 point for the correct $\ln(2)$ approximations and the comment on their accuracy.

2. Compare and contrast the following interpolation methods— ***Direct*** method, ***Newton's Divided Difference*** method, and ***Lagrangian*** method. 2 (CO3) (PO2)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x-x_0)(x-x_1)\ldots(x-x_{n-1})$	$f_n(x) = \sum_{i=0}^{n}\left(\prod_{j=0,j\neq i}^{n}\frac{x-x_j}{x_i-x_j}\right)f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3\log(\|\|A\|\| + \|\|b\|\|))$, or $O((n+1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can optimized further down to $O(n\log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 1 point for each valid comparison.

3. During a US presidential election campaign, polling analysts track how a candidate's support among likely voters responds to the amount of money spent on television advertising in a battleground state. Internal campaign data relate the candidate's projected support S (as a percentage of likely voters) to the cumulative advertising expenditure E (in millions of dollars). Table-1 portrays multiple recorded observations during the campaign. 4 (CO2) (PO1)

Table 1: Candidate support S as a function of advertising expenditure E for Question-3.

Expenditure, E (M$)	**Support,** S (%)
12.0	41.2
18.5	44.8
27.0	48.6
35.2	51.9

Determine the value of the candidate's support S when the advertising expenditure is $E = 23.4$ M$ using the Lagrangian method of interpolation and a second order polynomial.

Solution:

The 2nd order Lagrangian interpolant is —

$$\begin{aligned} S(E) &= \sum_{i=0}^{2} L_i(E) S(E_i) \\ &= L_0(E)S(E_0) + L_1(E)S(E_1) + L_2(E)S(E_2) \end{aligned}$$

Given, $E = 23.4$ \$M, and the 3 nearest data points that bracket it are, $E_0 = 18.5$, $E_1 = 27.0$, and $E_2 = 12.0$. The corresponding S values are $S(E_0) = 44.8$, $S(E_1) = 48.6$, and $S(E_2) = 41.2$ respectively. Now, let's calculate the Lagrangian weights.

$$L_0(E) = \prod_{\substack{j=0 \\ j\neq 0}}^{2} \frac{E - E_j}{E_0 - E_j} = \left(\frac{23.4 - 27.0}{18.5 - 27.0}\right) \times \left(\frac{23.4 - 12.0}{18.5 - 12.0}\right) = 0.743$$

$$L_1(E) = \prod_{\substack{j=0 \\ j\neq 1}}^{2} \frac{E - E_j}{E_1 - E_j} = \left(\frac{23.4 - 18.5}{27.0 - 18.5}\right) \times \left(\frac{23.4 - 12.0}{27.0 - 12.0}\right) = 0.438$$

$$L_2(E) = \prod_{\substack{j=0 \\ j\neq 2}}^{2} \frac{E - E_j}{E_2 - E_j} = \left(\frac{23.4 - 18.5}{12.0 - 18.5}\right) \times \left(\frac{23.4 - 27.0}{12.0 - 27.0}\right) = -0.181$$

Now that we have the weights, we can obtain the value of $S(23.4)$.

$$\begin{aligned} S(23.4) &= 0.743 \times S(18.5) + 0.438 \times S(27.0) + (-0.181) \times S(12.0) \\ &= 0.743 \times 44.8 + 0.438 \times 48.6 + (-0.181) \times 41.2 \\ &\approx 47.116\% \end{aligned}$$

Rubric:

- 1 point for the correct polynomial representation.
- 0.5 point for choosing the correct nearest points.
- 0.5 point for each Lagrangian weight $L_i(E)$ value.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 3 (CO4) (PO1)

Solution:

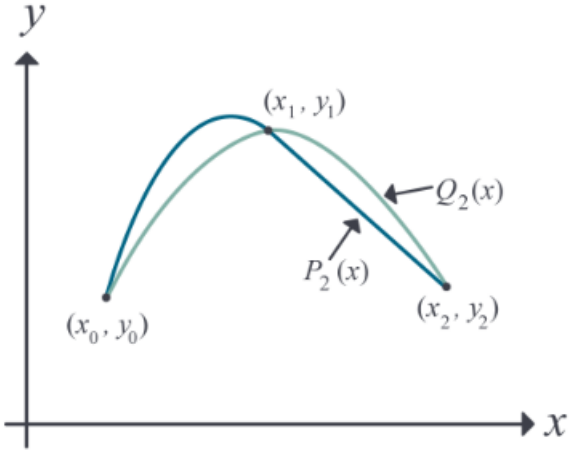


Figure 1: Interpolating polynomials through three points.

Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n+1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n+1$ data points (x_0, y_0), (x_1, y_1), $\ldots$, (x_n, y_n). Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n+1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$

Hence

$$R_n(x_i) = P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$
$$\Rightarrow R_n(x_i) = 0$$

The nth order polynomial $R_n(x)$ has $n+1$ zeros. A polynomial of order n can have $n+1$ zeros only if it is identical to a zero polynomial, that is,

$$R_n(x) \equiv 0$$
$$\Rightarrow P_n(x) \equiv Q_n(x)$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.